JUDGES' LIBRARY / DOSSIER 03

Technology & methodology.

An inspectable path from input to interpretation: rules, integration boundaries and reproducibility.

A deliberately simple architecture

Layer	Current implementation
Input	Local user entry or CSV validation; deterministic synthetic fixture in a separate demo workspace.
State	Client-side workspace and localStorage; no personal health submission or cloud sync endpoint.
Computation	Pure TypeScript functions normalize supported units, select prior days and apply PROHOR-IQR-3.0.
Presentation	React views show trends, data coverage, schematic atlas, NASA evidence and handover.
Public feeds	Same-origin read-only adapter for NASA OSDR metadata and NOAA Kp, with validation and fallback snapshots.
Offline	Vite build and service worker precache the app, public data and PDF library after a successful initial visit.

Trust boundary: public-source requests do not include personal readings. Local records are not submitted by the app to the public feeds. Browser extensions, hosting logs and exported files have separate privacy considerations.

Why keep the algorithm deterministic?

Given the same inputs and rule version, the same comparison can be reproduced without an internet connection or runtime AI service. The user can see the values and assumptions behind a result. Determinism improves inspectability; it does not establish medical validity.

02 / BASELINE RULE

The comparison, step by step.

Personal observations

For a local signal on day d, use days d−30 through d−1 inclusive; never include the current or future day. Select the latest timestamped reading for each prior day, with the last input row breaking exact ties. If that selected reading is invalid, it is excluded rather than replaced by an older same-day value. Missing and non-finite values are excluded; there is no zero filling.

A minimum of seven distinct usable prior days is required. Seven to 20 days are labeled provisional; 21 to 30 established. These are interface design choices, not statistical confidence or clinical certification.

Rule PROHOR-IQR-3.0
median = Q(0.50) · IQR = Q(0.75) − Q(0.25)
width = max(1.5 × IQR, signal-specific noise floor)
band = median ± width

Quantiles use linear interpolation at (n−1)q after sorting the values. A value strictly outside the applicable side of the band is marked for review. Equality is not flagged. The chart's shaded area is the middle 50% (IQR), not the wider change band; the exact band is in the explanation panel.

Worked synthetic example

For CDR-01 on flight day 146, the demo displays 5.6 hours of sleep against a 7.4-hour synthetic median. Sleep has a one-hour noise floor and the rule checks lower values. The example demonstrates interface behavior, not evidence that a crew member has a sleep disorder. The 30-day demo enrollment period is frozen at FD 111–140 for scenario days 142–147.

A stable but unhealthy baseline can still lead to a within-baseline state. Missing data are not a reassuring result; sudden or severe symptoms require approved procedures regardless of app state.

03 / DEPLOYMENT AND VERIFICATION

From source to a working build.

Public integration contract

The two allowlisted endpoints are **GET /api/feeds?feed=studies** and **GET /api/feeds?feed=space-weather**. The server checks upstream responses, applies a timeout and validates shapes. A request can return fresh network data, a prior server cache, or a bundled dated snapshot. The client also validates responses; a malformed successful response is not silently accepted.

Input quality and safety controls

Supported ranges and ordinal checks reject unsupported input; they are software checks rather than medical safety ranges. CSV imports validate as a batch, so invalid rows do not partially change the workspace. Local and demo modes do not share baseline records. A corrupted local workspace is protected from silent overwrite. Exported files are under user control and are not encrypted.

How to reproduce

Command	Purpose
npm ci	Install locked dependencies.
npm run typecheck	Check TypeScript project.
npm test	Run automated unit tests for engine, storage, feed and research behavior.
npm run build	Generate production assets and service-worker precache in web/.
npm start	Serve the built app and the same-origin public-feed adapter.

The repository includes browser checks for workflows, offline reload, mobile layouts, source-integrity behavior and selected accessibility rules. Passing software checks does not establish device safety, accessibility certification or clinical performance.

Sources & further reading

- Source repository
- Full methodology
- Architecture and security
- Test report